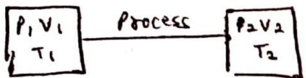
Kinetic Theory of Gases

Postulates of KTG :

1) Negligible size of gas molecules
2) Collisions are elastic
3) No force of interaction b/w particles
   i.e. PE = 0
4) Random / Brownian Motion
5) Instant connect, negligible time of contact.
6) No effect of gravity
7) $\boxed{PV = nRT}$ always valid.

State of a gas :

(P, V, T)



(For closed container) $\boxed{\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} = nR = \text{const.}}$

(*) Mixing of gases :→

Find P_f

$$\frac{P_1V_1}{T_0} + \frac{P_2V_2}{T_0} = \frac{P_f(V_1+V_2)}{T_0}$$

$$\Rightarrow \boxed{P_f = \frac{P_1V_1 + P_2V_2}{V_1+V_2}}$$

Modified gas law :

Container forms ⇒ $\boxed{PV = nRT}$ (n → No. of moles)

$\boxed{PV = NKT}$ N → Total no. of atoms

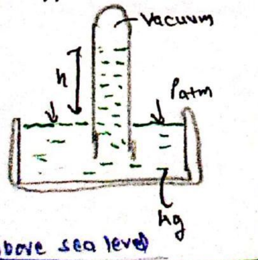
$(K = R/N_A)$

Atmospheric form ⇒ $\boxed{PM = dRT}$

M → Molar Mass
d → Density

Air Pressure & Barometric Relation :

→ Apparatus used to measure atm. pressure.



At sea level

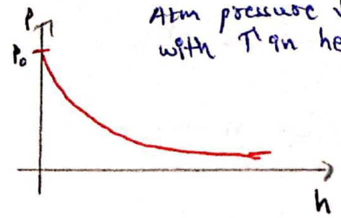
$h = 76\text{cm}$

$$\boxed{\begin{aligned} P_{atm} &= \rho_{Hg}\cdot g\cdot h \\ &= 1.01\times10^5\ Pa \\ &= 1\ atm \end{aligned}}$$

Above sea level

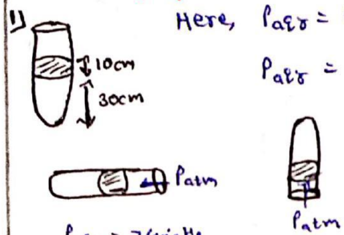
$\boxed{P = P_0 e^{\frac{-Mgh}{RT}}}$ P_0 → Atm press. at sea level
M → Molar mass of air

Atm pressure ↓ses exponentially with ↑ in height.



Cases of glass tube with Hg & Air :

1)



Here, $P_{air} = (76+10)$ cm Hg

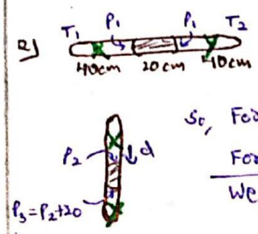
$P_{air} = 86$ cm of Hg

$P_{air} = 76$ cm Hg

$P_{air} = (76-10)$

$= 66$ cm Hg

2)



So, For X, $P_1 \times 40 = P_2 \times (40+d)$

For Y, $P_1 \times 40 = (P_2+20)(40-d)$

We can easily find d

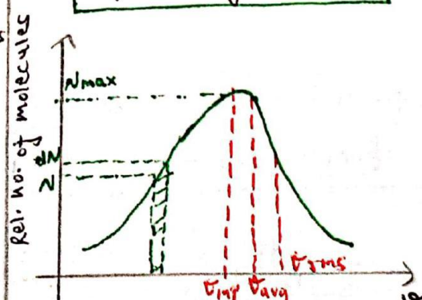
Molecular speed of gas molecules :

$$v_{avg} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8PV}{\pi nM}} = \sqrt{\frac{8P}{\pi d}}$$

$$v_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2PV}{nM}} = \sqrt{\frac{2P}{d}}$$

$$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3PV}{nM}} = \sqrt{\frac{3P}{d}}$$

$$\boxed{v_{mp} < v_{avg} < v_{rms}}$$

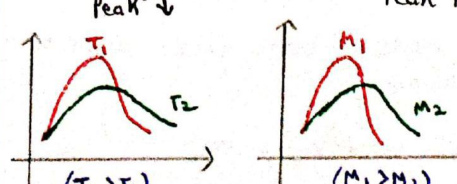


Maxwell's vel. distribution

→ $\boxed{v_{mp} \propto \sqrt{T}}$ & $\boxed{v_{mp} \propto \frac{1}{\sqrt{M}}}$

T↑ ⇒ Range ↑
Peak ↓

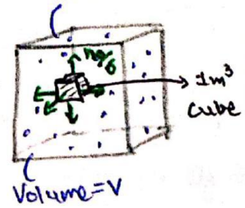
M↑ ⇒ Range ↓
Peak ↑



$(T_2 > T_1)$ $(M_1 > M_2)$

Pressure exerted by a gas :

Total Molecules = N



Lets assume,

Total no. of molecules $= N$

Molecular Density $(n_0) = \frac{N}{V}$

Mass of one molecule $= m'$

→ Per molecule space

$v' = \frac{V}{N}$

→ Density of gas

$\rho = m'\left(\frac{N}{V}\right) = m' n_0$

If we assume that the vibrations of gas molecules are equal in all dirn, then in each of 6 dirn (6 faces of cube), $(n_0/6)$ particle hit the wall of vessel.

→ No. of collisions / sec-m² on container wall

$$\boxed{\gamma = \frac{n_0}{6}\cdot v_{rms}}$$

→ Pressure can be written as change in momentum per second, per m²

$$P = 2m'v_{rms} \times \frac{n_0}{6} v_{rms}$$

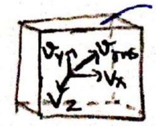
$$\Rightarrow \boxed{P = \frac{1}{3}\rho v_{rms}^2}$$

$$\Rightarrow \boxed{P = \frac{\rho RT}{M}}$$

[Atmospheric Gas law obtained when expression of v_{rms} is substituted]

Translational KE of a Gas :

Total molecules = N



For a single molecule

$$KE' = \frac{1}{2}m'v_x^2 + \frac{1}{2}m'v_y^2 + \frac{1}{2}m'v_z^2$$

$$KE' = \frac{1}{2} m' v_{rms}^2$$

$$KE' = \frac{1}{2} m' \left(\frac{3RT}{\underset{m' \cdot N_A}{M}}\right) = \frac{3}{2}\left(\frac{R}{N_A}\right)T$$

$$\Rightarrow \boxed{KE' = \frac{3}{2} KT}$$

Translational KE of single molecule

For whole Gas

$$\boxed{E = \frac{3}{2} NKT}$$ ($N \rightarrow$ Total no. of molecules)

$$\boxed{E = \frac{3}{2} nRT}$$ ($n \rightarrow$ Total moles of gas)

Equipartition of Energy & DOF :

→ Degree of freedoms are the dir^n in which gas molecule divides its energy equally.

- → Translational DOF → $f_T = 3$ (always)
- → Rotational DOF → $f_R = 2$ (Linear)
  $f_R = 3$ (Non-linear)
- → Vibrational DOF
  → (Value would be given in ques)

→ Law of equipartition of energy states that energy in each DOF is

$$\boxed{E = \frac{1}{2} kT}$$

For a gas molecule, Total energy

$$\boxed{E_T = \frac{f}{2} KT}$$

($f \rightarrow$ Total DOF for molecule)

For a Gaseous sample (n mol.)

$$\boxed{U = \frac{f}{2} nRT}$$ → Yahi hai, internal energy of a Gas

$$\Rightarrow \boxed{U = \frac{f}{2} PV}$$

DOF for a Gaseous Mixture :

gas-1 (f_1) → n_1 mol

gas-2 (f_2) → n_2 mol

⋮

gas-n (f_n) → n mol.

Then, for the gaseous mixture

$$\boxed{f_{eq} = \frac{f_1 n_1 + f_2 n_2 + \dots + f_n \cdot n}{n_1 + n_2 + \dots + n}}$$

⇓

$$f_{eq} = \left(\frac{\sum_{i=1}^{n} f_i n_i}{\sum_{i=1}^{n} n_i}\right)$$

Mixing of Gases at diff Temp :

Considering a sample of N-number of gases.

$$N\text{-Gases} \xrightarrow{\text{moles}} n_1, n_2 \dots n_N$$

$$\xrightarrow{\text{DOF}} f_1, f_2 \dots f_N$$

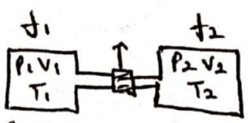
Temp → $T_1, T_2 \dots T_N$

→ Since all gases have diff. temp, due to temp. gradient heat transfer occurs within the gases in the isolated vessel.

→ Thus, at eq^m temp can be given by

$$\boxed{T_{eq} = \frac{f_1 n_1 T_1 + f_2 n_2 T_2 \dots + f_N n_N T_N}{f_1 n_1 + f_2 n_2 \dots + f_N n_N}}$$

Eg:



Sol^n

$$T_f = \frac{f_1 n_1 T_1 + f_2 n_2 T_2}{f_1 n_1 + f_2 n_2}$$

Also, moles of both gases would be conserved.

$$\Rightarrow n_1 + n_2 = n_f$$

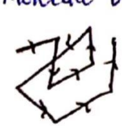
$$\left[\frac{P_1 V_1}{T_1} + \frac{P_2 V_2}{T_2} = \frac{P_f (V_1 + V_2)}{T_f}\right]$$

↓

Important approach for many questions

Mean free path :

It is the average distance travelled by a gas molecule between two successive collisions.



$$\text{Mean free path } (\lambda) = \frac{\text{Total distance in 1 sec}}{\text{Total collisions by one molecule in 1 sec}}$$

$$\Rightarrow \lambda = \frac{u_{avg}}{Z_1} = \frac{u_{avg}}{\sqrt{2}\pi\sigma^2 u_{avg} \times n}$$

$$\Rightarrow \boxed{\lambda = \frac{1}{\sqrt{2}\sigma^2 n^* \pi}}$$

where, $n^* \rightarrow$ No. of molecules / volume

$\sigma \rightarrow$ collision diameter (Diameter of molecule only)

Other form of λ :→

We know that,

$PV = nRT$

$PV = \frac{N}{N_A} RT$

$\Rightarrow \frac{N}{V} = \frac{P}{T}\left(\frac{N_A}{R}\right)$

$\frac{N}{V} = n^* = \frac{P}{KT}$

So,

$$\boxed{\cancel{\lambda = \frac{KT}{\sqrt{2}\pi\sigma^2 P}}}$$

$$\boxed{\lambda = \frac{KT}{\sqrt{2}\pi\sigma^2 P}}$$

Factors affecting λ :→

$$\left[\lambda \propto \frac{T}{P}\right]$$

1) At constant Temp, $P\uparrow \Rightarrow \lambda\downarrow$
2) At constant Pressure, $T\uparrow \Rightarrow \lambda\uparrow$
3) At constant volume, $P \propto T$, means (T/P) ratio is also constant, so no effect observed on changing P or T

Mean free Time (τ) :

It is average time b/w two successive collisions

$$\boxed{\tau = \lambda / u_{avg}}$$